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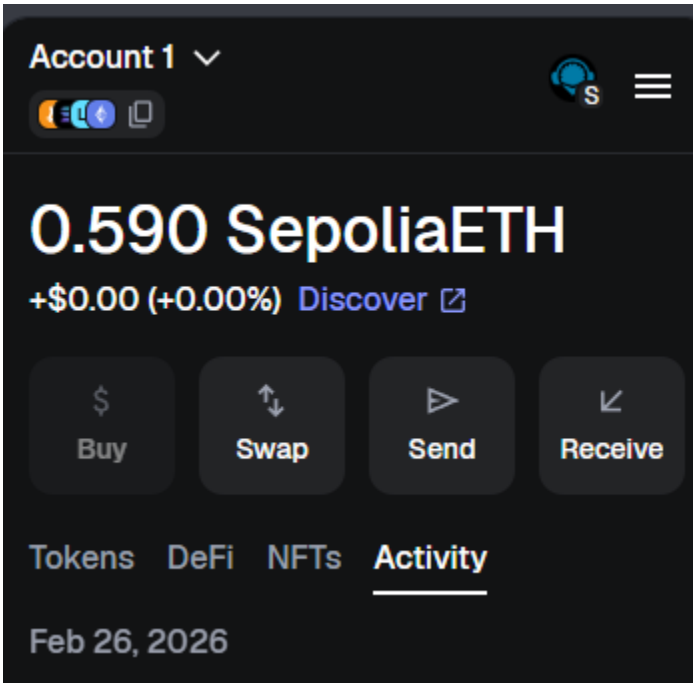
Requirements

- MetaMask installed
- Sepolia network enabled
- Sepolia ETH in **Account 1** (deployer) and ideally **Account 2** (receiver)
- Remix IDE: remix.ethereum.org
- Your ERC1155 contract code

Part A — MetaMask Sepolia Setup

1. MetaMask → enable **Test Networks**
2. Select **Sepolia**
3. Get Sepolia ETH from a faucet
4. Make sure you have enough for deployment + transactions

Checkpoint: You have Sepolia ETH in Account 1



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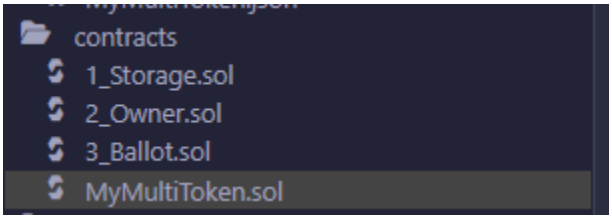
Part B — Remix Setup

1. Open Remix
2. Create file: MyMultiToken.sol
3. Paste your ERC-1155 contract code

Checkpoint: file saved in Remix



Part C — Fix OpenZeppelin Imports (if needed)

1. Go to **Solidity Compiler** → Compile once
2. If imports fail:
    - Add OpenZeppelin contracts using Remix Dependencies / Packages

Checkpoint: *no import errors*

Part D — Compile

1. Compiler version: **0.8.x** compatible with ^0.8.0
2. Click **Compile**

Checkpoint: *compilation success*

Part E — Connect Remix to MetaMask (Sepolia)

1. Deploy & Run Transactions tab
2. Environment: **Injected Provider (MetaMask)**
3. Approve MetaMask connection
4. Confirm network is **Sepolia**

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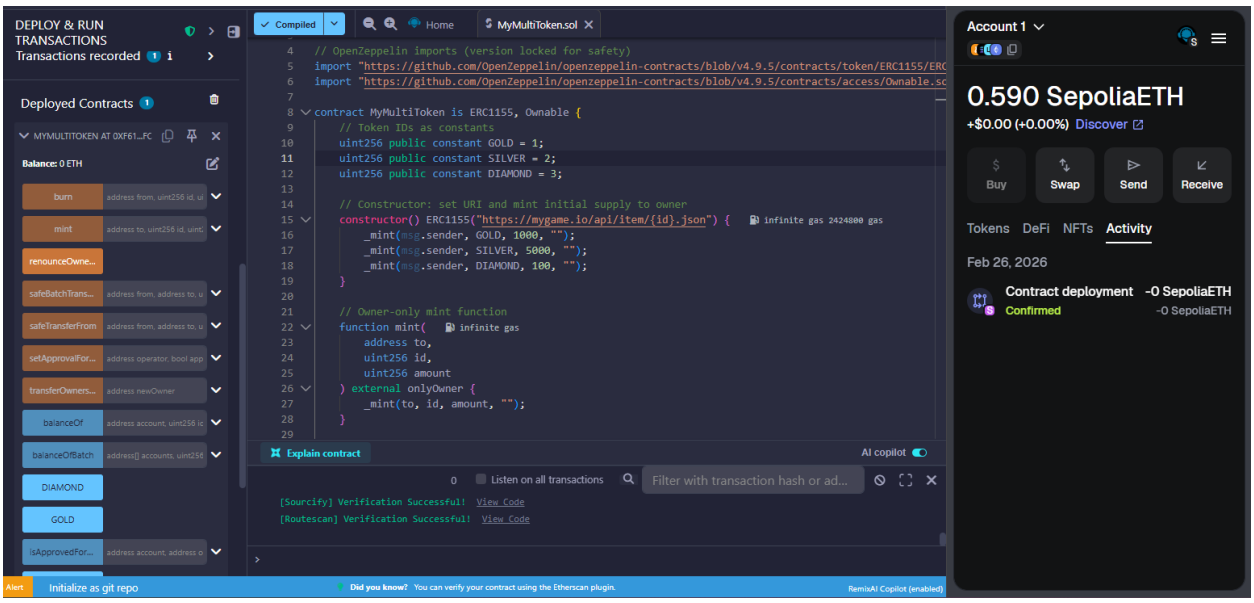
Checkpoint: *Remix shows your MetaMask account*

Part F — Deploy Contract

1. Select contract: MyMultiToken
2. Click **Deploy**
3. Confirm in MetaMask

Checkpoint: *contract appears under Deployed Contracts*



Part G — Read & Verify Initial Supplies

Test G1: Check initial balances (Account 1)

In the deployed contract UI, call:

- balanceOf(account1, 1) → should be **1000**
- balanceOf(account1, 2) → should be **5000**
- balanceOf(account1, 3) → should be **100**

Checkpoint: balances match the constructor mints

If your UI shows a balanceOf function:

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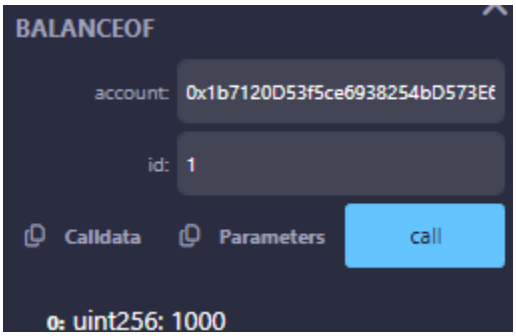
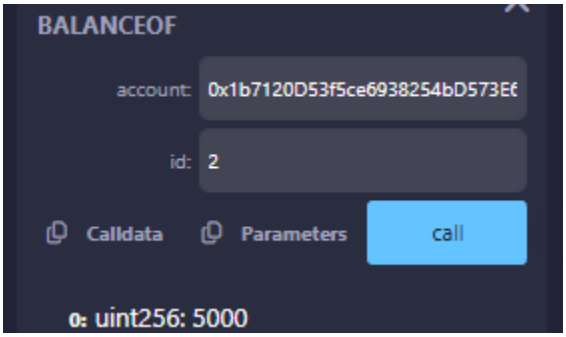
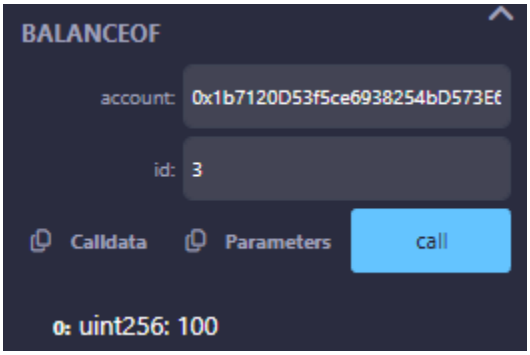
Enter:

- account: your address
- id: 1, 2, or 3







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Part H — Test Owner-only Mint

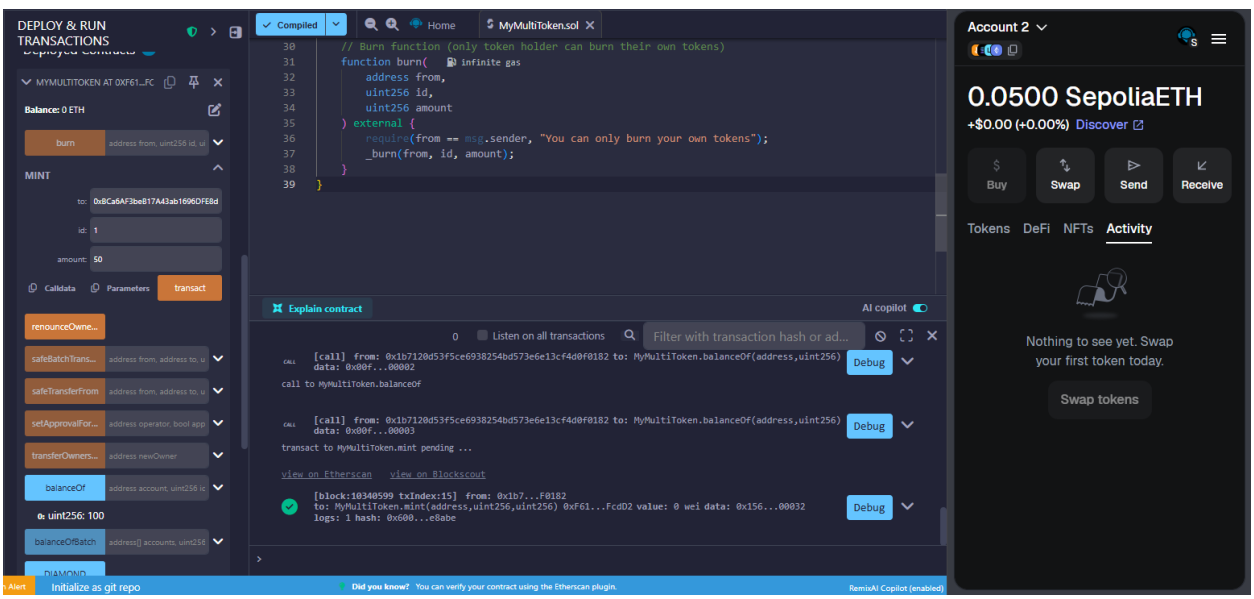
Test H1: Mint GOLD to Account 2 (owner action)

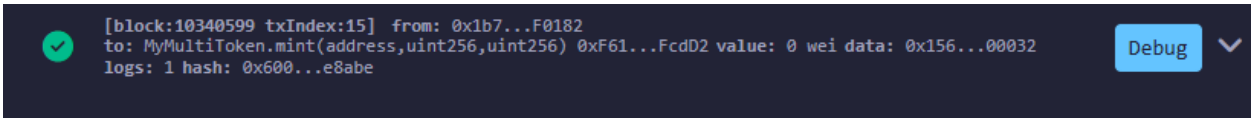
1. Switch nothing (stay in Account 1 owner)
2. Call: mint(to=Account2, id=1, amount=50)
3. Confirm transaction

Then verify:

- balanceOf(Account2, 1) → **50**
- balanceOf(Account1, 1) → still **1000** (because mint creates new tokens)

Checkpoint: *Account 2 received minted GOLD*





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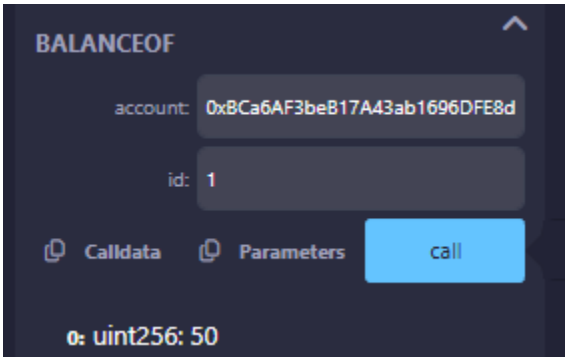
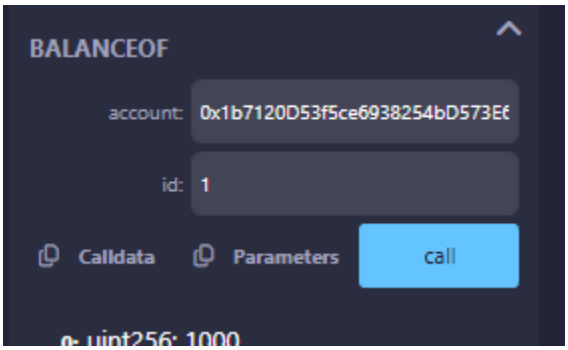
BALANCE OF ACCOUNT 2:



BALANCE OF ACCOUNT 1:



Test H2: Non-owner tries to mint (should FAIL)

1. Switch MetaMask to **Account 2**
2. In Remix, still Injected Provider (it follows MetaMask active account)
3. Call: mint(to=Account2, id=2, amount=10)

Expected: *Transaction **reverts** (fails) due to onlyOwner*

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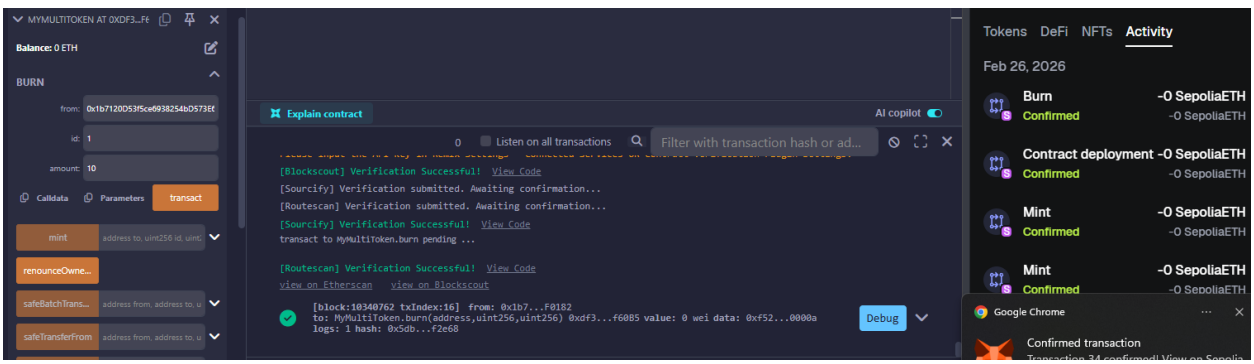
Part I — Burn Tests (Self-only)

Test I1: Burn your own tokens (Account 2)

1. Keep MetaMask on **Account 2**
2. Call: burn(from=Account2, id=1, amount=10)
3. Confirm transaction

Verify: balanceOf(Account2, 1) decreased from 50 → 40

Checkpoint: burn works for your own balance



Test I2: Attempt to burn someone else's tokens (should FAIL)

Still in Account 2:

- call burn(from=Account1, id=1, amount=1)

Expected: Revert with:

- **"You can only burn your own tokens"**

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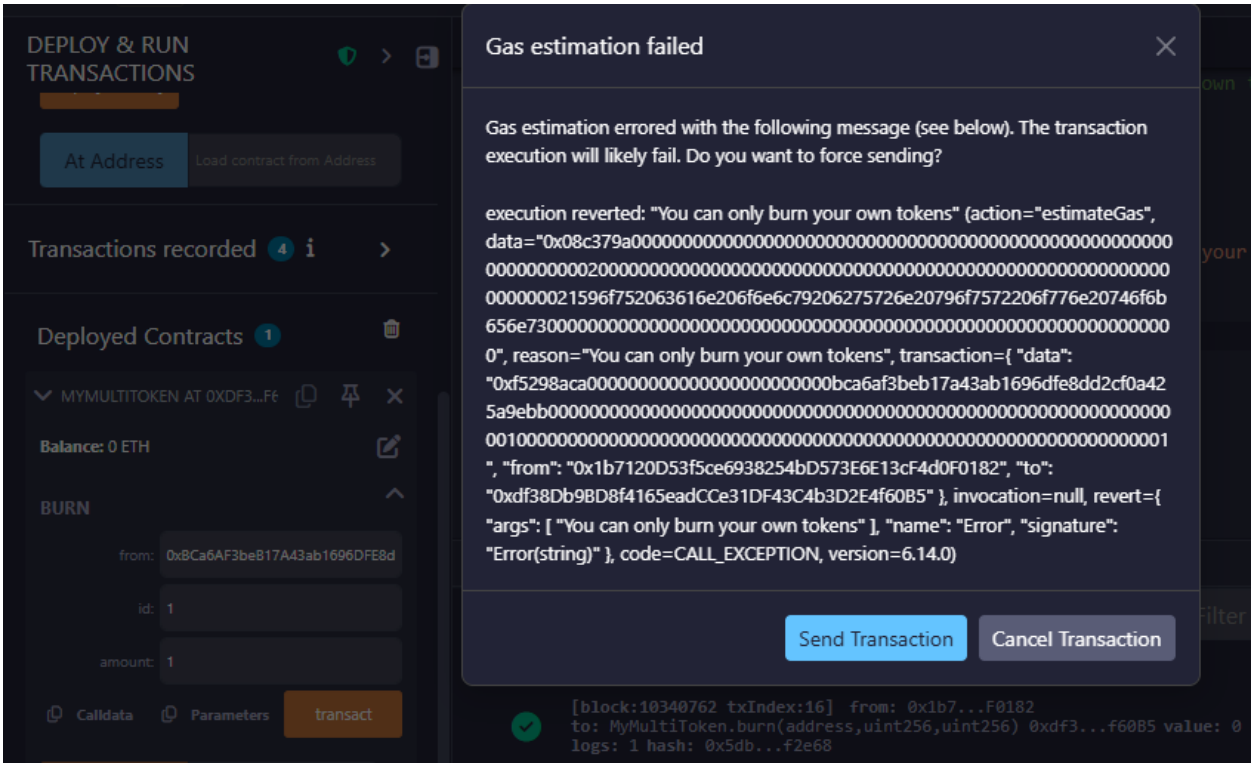

Part J — Transfer Tests (ERC-1155 Safe Transfer)

Test J1: Transfer GOLD from Account 1 to Account 2

1. Switch MetaMask back to **Account 1**
2. Use: safeTransferFrom(from=Account1, to=Account2, id=1, amount=25, data=0x)
3. Confirm transaction

Verify:

-
  -
    - balanceOf(Account2, 1) increases by 25
    - balanceOf(Account1, 1) decreases by 25

Checkpoint: *ERC1155 transfer works*

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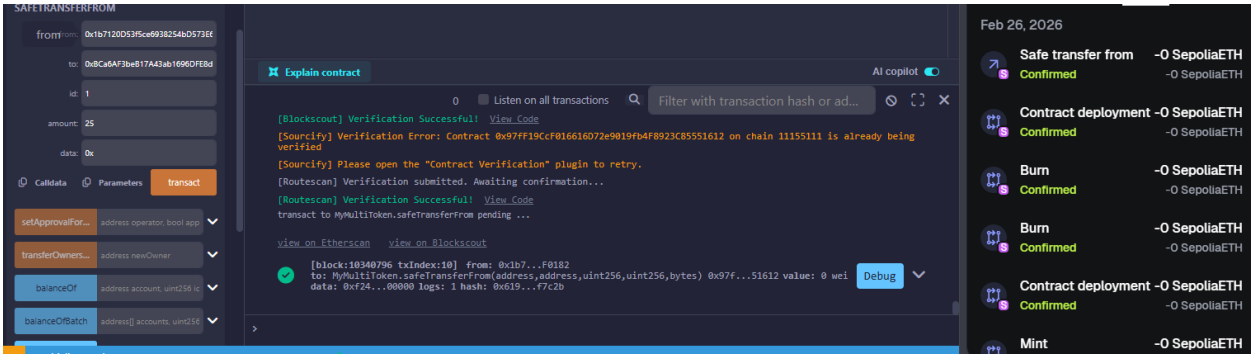
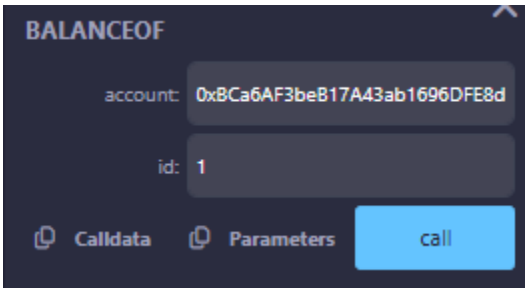
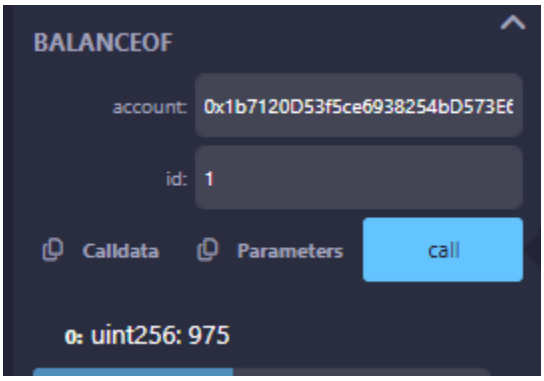


ACCOUNT 2:



ACCOUNT 1



Part K — Approval Tests (Operator Approval)

ERC-1155 supports "operators" who can transfer on your behalf:

- setApprovalForAll(operator, approved)
- isApprovedForAll(owner, operator)

Test K1: Approve Account 2 as operator for Account 1

1. MetaMask: Account 1

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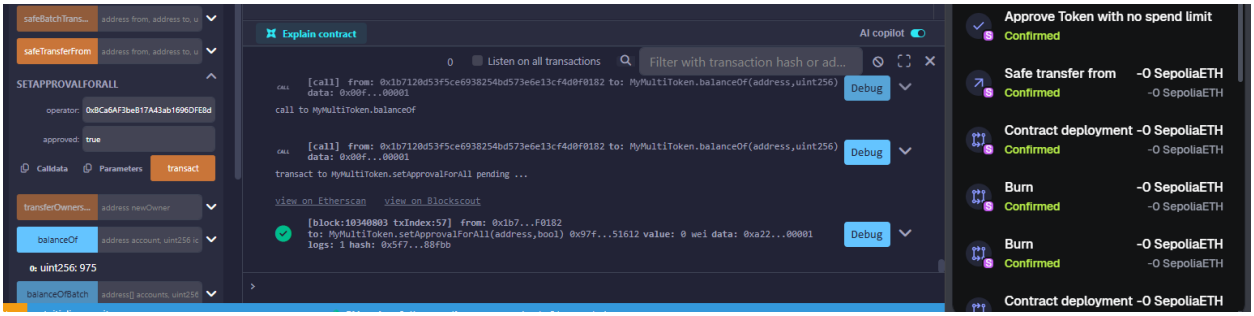
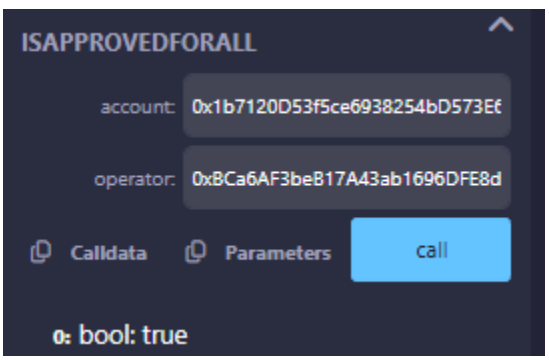
2. Call:
    - setApprovalForAll(Account2, true)
3. Verify:
    - isApprovedForAll(Account1, Account2) → **true**

Checkpoint: operator approval enabled





Test K2: Transfer from Account 1 using Account 2 as operator

1. Switch MetaMask to Account 2
2. Call:
    - safeTransferFrom(from=Account1, to=Account2, id=2, amount=100, data=0x)

Expected: should succeed **only if approval is true**

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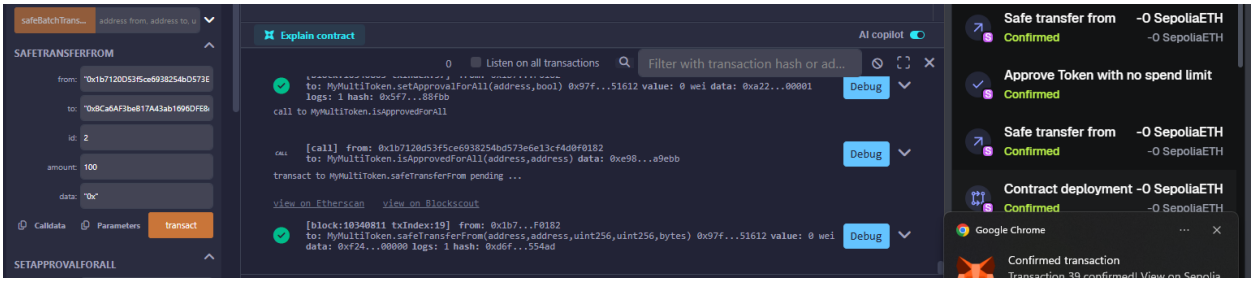

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4) Common gotchas (ERC-1155 specific)

- **URI doesn't open**: your URI is a sample domain; it won't resolve unless you host metadata.
- **{id} formatting**: ERC-1155 expects {id} as 64-hex padded by many implementations; OpenZeppelin follows the standard substitution behavior.
- **Burn restriction**: your burn requires from == msg.sender — operators cannot burn on behalf (unless you change logic).
- **Transfers require data bytes**: in Remix, use 0x or 0x00... as needed.

---

5) Post-lab questions (ERC-1155 version)

1. What are the token IDs for GOLD, SILVER, DIAMOND?

   uint256 public constant GOLD = 1;

   uint256 public constant SILVER = 2;

   uint256 public constant DIAMOND = 3;

2. Why can Account 2 not call mint()?

   Account 2 cannot call mint() because the function is restricted by the onlyOwner modifier. Only the contract owner (Account 1, the deployer) is allowed to mint new tokens, so any attempt by Account 2 will revert.

3. What function confirms how many tokens of an ID a wallet owns?

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The function that confirms how many tokens of a specific ID a wallet owns is balanceOf(address account, uint256 id). It requires both the wallet address and the token ID to return the balance.

4. Why does burn(Account1, id, amount) fail when called by Account 2?

   The call burn(Account1, id, amount) fails when executed by Account 2 because the burn function requires that from must be equal to msg.sender. This means users can only burn their own tokens, not someone else's.

5. What is the difference between _mint() and safeTransferFrom()?

   The difference between _mint() and safeTransferFrom() is that _mint() creates new tokens and increases supply, while safeTransferFrom() transfers existing tokens from one address to another without creating new tokens.